



B(5th Sm.)-Computer Science-G/DSE-A-1, DSE-A-2 &
DSE-A-3/CBCS

2024

COMPUTER SCIENCE — GENERAL

Paper : DSE-A-1, DSE-A-2 and DSE-A-3

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Paper : DSE-A-1

(Database Management System)

Full Marks : 50

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions :

2×5

- What do you mean by data redundancy?
- What do you mean by full functional dependency?
- Explain transitivity rule of functional dependency with an example.
- What do you mean by instance and schema?
- What is multi-valued attribute? Give an example.
- What do you mean by meta data?
- Define : 3NF.
- Mention two problems of Unnormalized Database.

2. (a) Draw an ER diagram for Hospital Management System by clearly specifying

- Entity names
- Relationship names
- Mapping Cardinalities
- Attributes and Primary Key.

(b) Explain the various types of integrity constraints with proper examples.

6+4

3. (a) Describe the three layer architecture of DBMS.

(b) Explain the difference between physical and logical data independence.

(c) State the difference between schema and instance.

5+3+2

4. (a) Consider the relational schema $R(A, B, C, G, H, I)$ on which following FDs hold :

$F = \{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$, find the closure of the FDs.

(b) What is weak entity set? Explain with an example.

(c) What is DML?

6+2+2

Please Turn Over

(0785+0786+0787)

5. (a) Consider the tables given below and write the following SQL queries :
- Employee (Empid, Ename, Salary, Age)
Dept (Deptid, Dname, Dlocation)
- (i) Find the name of the employee who gets the highest salary.
(ii) Find the name of the employees who are working in 'CS' department and have salary greater than 5000 rupees.
(iii) Find the age of the employee who works in a department located in 'Kolkata'.
- (b) "BCNF is stronger than 3NF."— Verify the statement with proper justification. (2+2+2)+4
6. (a) Define : BCNF.
(b) Given a relation R(A, B, C, D, E) and Functional Dependency set $FD = \{AB \rightarrow C, D \rightarrow E\}$, determine whether the given R is in 2NF, with proper justification. If not, convert it into 2NF.
(c) Discuss about different cardinality of ER model. 2+4+4
7. (a) Differentiate between super key and candidate key with example.
(b) Define ternary relationship. Give example.
(c) Distinguish between Lossless and Lossy join decomposition. Give examples. 3+(2+1)+(2+2)
8. (a) Why are Armstrong's Axioms considered sound and complete with respect to functional dependencies? Explain.
(b) Explain the concept of augmentation rule in Armstrong's Axioms. How this rule assist in deriving new functional dependencies? Explain with suitable example.
(c) What is the function of division operation in relational algebra? 3+(2+3)+2